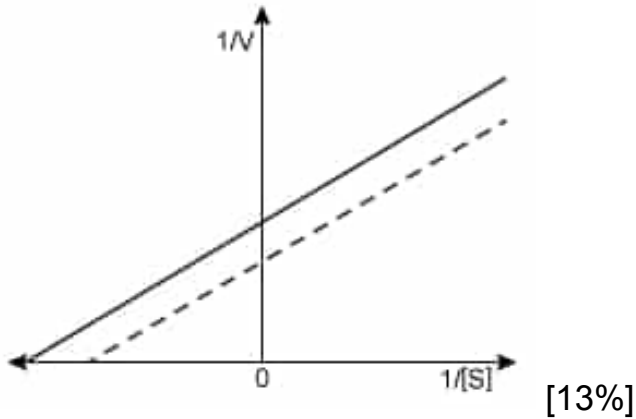
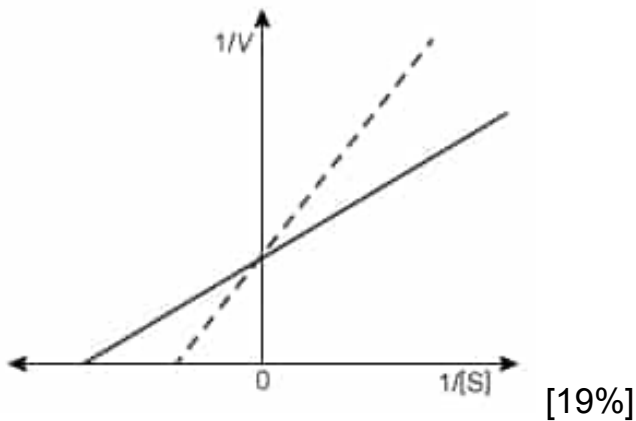


Interleukin-1 β is a cytokine that upregulates the synthesis of PGE₁, the final enzyme in the arachidonic acid pathway that generates prostaglandin E. Which of the following plots most likely represents the change in the kinetics of prostaglandin E production from cellular lysates obtained from cells incubated with (dashed line) and without (solid line) interleukin-1 β ?

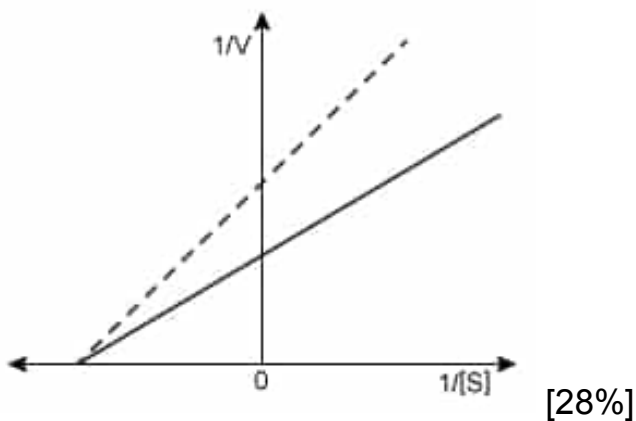
☐ A.



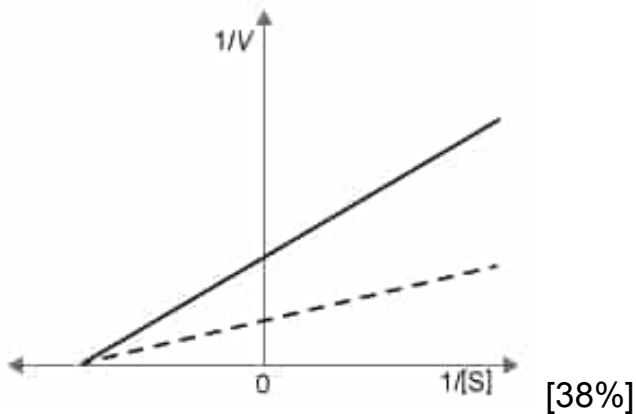
☐ B.



☐ C.



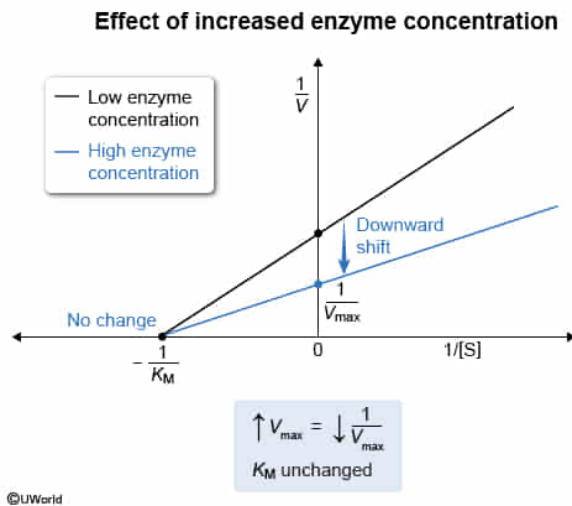
✓ ☒ D.



Correct

37% answered correctly

Explanation:



Lineweaver-Burk plots are used to determine Michaelis-Menten kinetic parameters. These **linear plots** relate the inverse of reaction velocity ($1/V$) on the y-axis to the inverse of substrate concentration ($1/[S]$) on the x-axis. This linear relationship is based on **rearrangements** of the **Michaelis-Menten equation**.

On a Lineweaver-Burk plot, the x-intercept corresponds to the negative inverse of the Michaelis constant (x-intercept = $-1/K_M$) and the y-intercept corresponds to the inverse of the maximum velocity (y-intercept = $1/V_{\max}$). Because the y-intercept in a Lineweaver-Burk plot is $1/V_{\max}$, an increase in $V_{\max}$ would correspond to a **lower y-intercept value**. The enzyme affinity for substrate (K_M) is independent of enzyme concentration, so the x-intercept ($-1/K_M$) would remain unchanged with varying enzyme concentration.

In this example, interleukin- 1β upregulates the synthesis of the enzyme PGE1, which results in **increased enzyme concentration**. The maximum velocity $V_{\max}$ of the enzyme-catalyzed reaction, which occurs when the enzyme is saturated with substrate, is equal to the product of the catalytic rate constant k_{cat} and the concentration of enzyme in solution:

$$V_{\max} = k_{\text{cat}}[E_{\text{tot}}]$$

Therefore, increasing the enzyme concentration would result in an **increase in the maximum velocity** of the reaction and a decrease in the y-intercept value.

(Choice A) This plot correctly shows an increase in $V_{\max}$ but also demonstrates a change in K_M . An enzyme's K_M is unchanged by changes in concentration.

(Choice B) This plot represents an enzyme system in the presence of a competitive inhibitor, in which $V_{\max}$ is not changed.

(Choice C) This plot represents an enzyme in the presence of a noncompetitive inhibitor, in which K_M is unchanged but $V_{\max}$ decreases.

Educational objective:

in Michaelis-Menten kinetics, an increase in the amount of enzyme available in a particular system causes an increase in $V_{\max}$ but no change in K_M . Lineweaver-Burk graphs are linear plots that relate inverse reaction velocity ($1/V$) to inverse substrate concentration ($1/[S]$). In a Lineweaver-Burk plot, an increase in enzyme concentration causes a decrease in the y-intercept ($1/V_{\max}$) but no change in the x-intercept ($-1/K_M$).

Time spent:0 sec

Subject:
Biochemistry

QID:400757

System:
Enzymes